Statistics 280 — Practice Midterm 2A — Solutions

Point values per part match the exam. Normal-table values follow the table at the back of the exam. R answers were run in R 4.3 (the ## lines are real output); any equivalent correct R command earns full credit.

Q.1 (30)

a (5 points)
2 of the 12 keys work: $2/12 = 1/6 \approx 0.17$.

b (6 points)
The first two fail and the third works: $\frac{10}{12} \cdot \frac{9}{11} \cdot \frac{2}{10} = \frac{3}{22} \approx 0.14$.

c (6 points)
More than 4 tries means the first 4 keys all fail: $\frac{10}{12} \cdot \frac{9}{11} \cdot \frac{8}{10} \cdot \frac{7}{9} = \frac{14}{33} \approx 0.42$. Equivalently $\binom{10}{4} / \binom{12}{4} = 210/495$.

d (7 points)
The first 5 keys are a random subset of 5 from 12. Both working keys must be in it, along with 3 of the 10 bad keys: $\binom{2}{2}\binom{10}{3} / \binom{12}{5} = 120/792 \approx 0.15$. (It's a hypergeometric probability.)

e (6 points)
$2/12 = 1/6$. By symmetry, every key is equally likely to be in position 7, just as the second card dealt is an ace with chance 4/52 (Lecture 9). The order you try keys in doesn't change the unconditional chance at any single position. It only changes conditional chances, such as the chance given that the earlier keys failed.

Q.2 (30)

a (6 points)
Maximum: $\min(160, 150) =$ **150**, when every CS student also takes statistics. Minimum: $160 + 150 - 200 =$ **110**, when the overlap is as small as it can be, which is when every resident takes at least one of the two. In probabilities: max $= \min(P(S), P(C))$, min $= \max(P(S) + P(C) - 1, 0)$.

b (6 points)
$P(S \text{ and } C) = 0.8 \times 0.75 = 0.6$, so **120** residents. $P(S \text{ or } C) = 0.8 + 0.75 - 0.6 = 0.95$, so **190** residents.

c (8 points)
Maximum: **140**, the smallest group, when all of I is inside both S and C. Minimum: at most 40 residents are not in S, 50 are not in C, and 60 are not in I. So at most $40 + 50 + 60 = 150$ residents miss at least one group, and at least $200 - 150 =$ **50** are in all three. It is attained when those three "missing" groups don't overlap. In general: $P(S \cap C \cap I) \geq P(S) + P(C) + P(I) - 2$ (Review 9–10 Q8).

d (4 points)
No. From a), at least 110 residents are in both, so $P(S \text{ and } C) > 0$. In general, independent and mutually exclusive would mean $0 = P(A \text{ and } B) = P(A)P(B)$, which forces $P(A) = 0$ or $P(B) = 0$.

e (6 points)
$P(C|S) = 112/160 = 0.70$ and $P(S|C) = 112/150 \approx 0.75$. They're **not independent**: $P(C|S) = 0.70 \neq$

$P(C) = 0.75$. (Equivalently, 112 ≠ 120.) Frequency interpretation: if we repeatedly chose a resident at random and kept only the times we got a statistics student, about 70% of those would also be taking CS.

Q.3 (30)

a (4 points)
$0.2/0.8 = 1/4$.

b (4 points)
$P(\text{winner}|\text{spam})/P(\text{winner}|\text{legit}) = 0.40/0.02 = 20$.

c (6 points)
Posterior odds = prior odds × data odds $= \frac{1}{4} \times 20 = 5$. Probability $= 5/(1+5) = 5/6 \approx 0.83$. Check with Bayes: $0.2 \cdot 0.4/(0.2 \cdot 0.4 + 0.8 \cdot 0.02) = 0.08/0.096$.

d (6 points)
With the stated independence, the time of sending contributes its own data odds, $0.30/0.05 = 6$. Posterior odds $= \frac{1}{4} \times 20 \times 6 = 30$, so the probability is $30/31 \approx 0.97$.

e (5 points)
We need posterior odds 1, so prior odds $\times\, 20 = 1$, which gives prior odds $1/20$. Prior probability $= \frac{1/20}{1+1/20} = 1/21 \approx$ **4.76%**.

f (5 points)
Law of total probability: $0.2(0.40) + 0.8(0.02) = 0.096$, so **9.6%**. Interpretation of c): among many emails that contain "winner", about 5 in 6 are spam in the long run.

Q.4 (40)

a (8 points)
Hypergeometric: $\frac{\binom{12}{2}\binom{28}{10}}{\binom{40}{12}} \approx 0.155$.

```
choose(12, 2) * choose(28, 10) / choose(40, 12)
```

```
## [1] 0.1550292
```

b (4 points)

```
dhyper(2, 12, 28, 12)
```

```
## [1] 0.1550292
```

Order: successes drawn, success tickets (12 under 30), failure tickets (28), number drawn (12).

c (4 points)

```
phyper(1, 12, 28, 12)
```

```
## [1] 0.05156962
```

d (5 points)
At least 4 = 1 − P(at most 3).

```
1 - phyper(3, 12, 28, 12)
```

```
## [1] 0.5214202
```

e (7 points)
Binomial: $\binom{12}{2}(0.3)^2(0.7)^{10} = 66 \times 0.09 \times 0.0282 \approx 0.168$.

```
dbinom(2, 12, 0.3)
```

```
## [1] 0.1677903
```

f (4 points)

```
pbinom(5, 12, 0.3) - pbinom(1, 12, 0.3)
```

```
## [1] 0.7971262
```

Or `sum(dbinom(2:5, 12, 0.3))`.

g (4 points)
$EV = 12 \times 0.3 = 3.6$, $SE = \sqrt{12 \times 0.3 \times 0.7} \approx 1.59$.

h (4 points)
In a), the jury is drawn *without replacement* from a small pool, so the draws are dependent: once a young juror is chosen, the chance the next one is young drops to 11/39. The binomial needs independent draws with the same success probability each time.

Q.5 (30)

a (8 points)
P(pay 3) = 2/6 = 12/36. P(win 1) = 3/6 + (1/6)(1/6) = 18/36 + 1/36 = 19/36, because a 6 then a 1 also wins $1. Each of $2, $3, $4, $5, $6 has chance (1/6)(1/6) = 1/36.

x	−3	1	2	3	4	5	6
P(X=x)	12/36	19/36	1/36	1/36	1/36	1/36	1/36

Check: the probabilities sum to 36/36.

b (8 points)
$E(X) = \frac{-36+19+(2+3+4+5+6)}{36} = \frac{3}{36} = \frac{1}{12} \approx 0.08$. $E(X^2) = \frac{9 \cdot 12+19+(4+9+16+25+36)}{36} = \frac{217}{36} \approx 6.03$.
$SE(X) = \sqrt{217/36 - (1/12)^2} \approx$ **2.45**.

c (7 points)
One solution among many:

```
play = function() {
  roll = sample(1:6, 1)
  if (roll <= 2) {
    return(-3)
  } else if (roll <= 5) {
    return(1)
  } else {
    return(sample(1:6, 1))
  }
}
```

d (7 points)

```
set.seed(2)
N = 100000
x = numeric(N)
for (i in 1:N) {
  x[i] = play()
}
mean(x)                       # estimates E(X) = 1/12
sd(x) * sqrt((N - 1) / N)     # estimates SE(X) = 2.45
mean(x > 0)                   # estimates P(X > 0) = 24/36
```

```
## [1] 0.09405
## [1] 2.45885
## [1] 0.66809
```

Q.6 (40)

a (6 points)
The box has 10 tickets: 8 marked 1 (make) and 2 marked 0 (miss). Average 0.8, SD $\sqrt{0.8 \times 0.2} = 0.4$. Draw 400 times with replacement. EV $= 400 \times 0.8 = 320$, SE $= \sqrt{400} \times 0.4 = 8$.

b (5 points)
EV $= 80\%$, SE $= 0.4/\sqrt{400} = 0.02 = 2\%$.

c (5 points)

```
1 - pbinom(329, 400, 0.8)
```

```
## [1] 0.1163917
```

d (10 points)
At least 330 means 329.5 or more with the continuity correction. $z = (329.5 - 320)/8 = 1.19$. The table gives an area of about 76.6% between −1.19 and 1.19, so the right tail is about $(100 - 76.6)/2 \approx$ **11.7%**. (The exact answer from c) is 11.6%.)

e (6 points)
The **100-shot stretch**. The SE of the percentage is $0.4/\sqrt{n}$: 4% for 100 shots and 2% for 400. 85% is 1.25 SEs above 80% for 100 shots but 2.5 SEs above for 400. With more shots, the percentage settles closer to 80% (the law of averages), even though the absolute chance error in the number made grows.

f (8 points)
Box: tickets +1 (80%) and −3 (20%). Average $= 0.8 - 0.6 = \$0.20$. SD $= (b - s)\sqrt{p_b p_s} = 4\sqrt{0.8 \times 0.2} =$

$\$1.60$. Over 400 shots: EV $= \$80$, SE $= \sqrt{400} \times 1.6 = \$32$. Losing money means net < 0: $z = (0 - 80)/32 = -2.5$, so the chance is about $(100 - 98.76)/2 \approx$ **0.62%**.

Q.7 (15 extra credit)

A splits into the mutually exclusive pieces "A and B" and "A and not-B", so $P(A) = P(A \text{ and } B) + P(A \text{ and not-}B)$. Then

$$P(A \text{ and not-}B) = P(A) - P(A \text{ and } B) = P(A) - P(A)P(B) = P(A)\,(1 - P(B)) = P(A)\,P(\text{not-}B),$$

using independence in the second step. That is exactly the definition of independence for A and not-B.